

COMPREHENSIVE 200 QUESTION AND ANSWER ON BLY 121 FOR EXAM GUIDE BY **DAVID OCHE DESIGN**

DISCLAIMER: This question and answer are maint to prepare we the students for Bly 121 examination, so therefore it is for an academic purposes.

1. _____ are classified according to morphological and developmental characteristic, such as a body plan. (a) Human (b) tree **(c) Animal** (d) plant
2. The animal body plan is _____ **(symmetrical)**
3. Animals may be asymmetrical, radial or bilateral in form. **TRUE OR FALSE**
4. _____ animals are animals with no pattern or symmetry. **(Asymmetrical)**
5. an example of an asymmetrical animal is a _____ **(sponge)**
6. An organism with _____ has a longitudinal (up-and down) orientation **(radial symmetry)**
7. An example of an organism with radial symmetry is a **(sea anemone)**
8. Most animal species undergo a layering of early tissues during embryonic development. These layers are called **(germ layers)**
9. Each layer develops into a specific set of _____ and _____ **(tissue and organs)**
10. The animals that display radial symmetry develop two germ layers, which are _____ and _____ **(an inner layer (endoderm) and an outer layer (ectoderm)).**
11. Animals with three germ layers are called. **(Triploblast)**
12. Animals with two germ layers are called. **(Diploblast)**
13. Animals with bilateral symmetry develop three germ layers which are _____, _____ and _____ **(an inner layer (endoderm),an outer layer (ectoderm), and a middle layer (mesoderm).)**
14. During embryogenesis, diploblasts develop two embryonic germ layers **(an ectoderm and an endoderm.).**
15. Triploblasts may develop an internal body cavity derived from mesoderm, called a _____ **(coelom)**

16. The epithelial-lined cavity with a space, usually filled with fluid, which lies between the digestive system and the body wall is called. (Coelom)
17. Triploblasts that do not develop a coelom are called. (acoelomates)
18. Examples of acoelomates include the _____ (flatworms)
19. Animals with a true coelom are called _____
(eucoelomates or coelomate)
20. Examples of acoelomates include the _____ (flatworms)
21. Animals such as earthworms, snails, insects, starfish, and vertebrates are all _____
(eucoelomates)
22. A third group of triploblasts has a body cavity that is derived partly from mesoderm and partly from endoderm tissue. These animals are called (pseudocoelomates)
23. _____ are example of pseudocoelomates. (roundworms)
24. bilaterally symmetrical, triploblastic eucoelomates can be divided into two groups based on differences in their early embryonic development. they include (protostomes and deuterostomes)
25. The word protostome comes from Greek words meaning (mouth first)
26. The word deuterostomes comes from Greek words meaning (mouth second)
27. The difference between protostomes and deuterostomes reflects the fate of a structure called the (blastopore)
28. Animals in subkingdom Parazoa represent the simplest animals and include the (sponges, or phylum Porifera)
29. The body of the simplest sponges takes the shape of a cylinder with a large central cavity called the (spongocoel)
30. Water enters the spongocoel from numerous pores in the body wall. Water flows out through a large opening called the (osculum)
31. Sponges consist of an outer layer of flattened cells and an inner layer of cells called (choanocytes) separated by a jelly-like substance called (mesohyl)

32. The mesohyl contains embedded amoeboid cells that secrete tiny needles called (spicules)
33. In sponges, Food particles are trapped in mucus produced by the sieve-like collar of the choanocytes and are ingested by (phagocytosis).
34. _____ take up nutrients repackaged in food vacuoles of the choanocytes and deliver them to other cells within the sponge. (Amoebocytes)
35. Sponges reproduce both _____ and _____ (sexually and asexually)
36. The asexual reproduction in sponges is either by. (Fragmentation or budding)
37. A type of asexual reproduction found only in fresh water sponges occurs through the formation of _____ (gemmules)
38. Sponges are monoecious (or hermaphroditic). TRUE OR FALSE
39. Sponges Eggs arise from amoebocytes and are retained within the (spongocoel)
40. Sponges sperm arise from choanocytes and are ejected through the (osculum).
41. The phylum _____ includes animals that show radial or biradial symmetry and are diploblastic. (cnidarian)
42. Cnidarians have specialized cells known as (cnidocytes) containing organelles called (nematocysts.)
43. Cnidarians display two distinct body plans which are. (polyp or "stalk" and medusa or "bell")
44. The best-known medusoid animals are the jellies (jellyfish)
45. All cnidarians have two tissue layers. The outer layer is called the (epidermis), whereas the inner layer is called the (gastrodermis)
46. Cnidarian cells exchange oxygen, carbon dioxide, and nitrogenous wastes by _____ between cells in the epidermis and gastrodermis with water.(diffusion)
47. The phylum Cnidaria contains about 10,000 described species divided into four classes which are (Anthozoa, Scyphozoa, Cubozoa, and Hydrozoa).
48. The _____ & are acoelomate organisms that include free-living and parasitic forms. (flatworms)

49. The nematodes, or roundworms, possess a _____ and consist of both free living and parasitic forms. (pseudocoelom)
50. the arthropods, one of the most successful taxonomic groups on the planet, are _____ organisms with a hard exoskeleton and jointed appendages. (coelomate)
51. The nematodes and the arthropods belong to a clade with a common ancestor, called (Ecdysozoa).
52. _____ refers to the periodic shedding, or molting, of the exoskeleton. (Ecdysis)
53. The _____ is a single layer of cells or a layer of fused cells covering a layer of circular muscle above a layer of longitudinal muscle. (epidermal tissue)
54. Flatworms are traditionally divided into four classes which are (Turbellaria, Monogenea, Trematoda, and Cestod)
55. _____ are responsible for serious human diseases including schistosomiasis, caused by a blood fluke (Schistosoma). (Trematodes)
56. The remaining body of the tapeworm is made up of a long series of units called (proglottids)
57. The name Nematoda is derived from the Greek word “_____,” which means “thread.” (nemos)
58. In nematodes, Nitrogenous wastes are removed by (diffusion)
59. Nematodes may reproduce asexually by (parthenogenesis)
60. The name “arthropoda” means (jointed legs)
61. _____ dominate the animal kingdom with an estimated 85 percent of known species (Arthropoda)
62. The insects we encounter on a daily basis— such as ants, cockroaches, butterflies, and bees—are examples of (Hexapoda)
63. _____ includes animals such as spiders, scorpions, horseshoe crabs, and sea spiders. (Subphylum Chelicerata)
64. The phyla Mollusca and Annelida belong to a clade called the (Lophotrochozoa)

65. _____ is the predominant phylum in marine environments, where it is estimated that 23 percent of all known marine species belong to this phylum. (Mollusca)
66. many mollusks have a scraping structure at the mouth, called a (radula)
67. Animals in the class Polyplacophora (“bearing many plates”) are commonly known as (chitons)
68. Phylum _____ are segmented worms found in marine, terrestrial, and freshwater habitats. (Annelida)
69. Annelids segmented body plan results in repetition of internal and external features in each body segment. This type of body plan is called (metamerism).
70. Excretion in Annelids is carried out by pairs of primitive “kidneys” called _____ that consist of a convoluted tubule and an open, ciliated funnel present in every segment. (metanephridia)
71. Earthworms are the most abundant members of the subclass Oligochaeta, distinguished by the presence of the _____ (clitellum)
72. _____ is a ring structure in the skin of earthworm that secretes mucus to bind mating individuals and forms a protective cocoon for the eggs. (clitellum)
73. The chaetae of polychaetes are also arranged within fleshy, flat, paired appendages on each segment called (parapodia).
74. Significant differences between leeches and other annelids include the development of (suckers) at the anterior and posterior ends, and the absence of (chaetae).
75. _____ include the phyla Echinodermata and Chordata (which includes the vertebrates) and two smaller phyla. (Deuterostomes)
76. Echinoderms have a unique system for gas exchange, nutrient circulation, and locomotion called (the water vascular system)
77. Different members of Echinodermata include the following, except (a) sea star in class Asteroidea, (b) the brittle star in class Ophiuroidea, (c) the sea urchins of class Echinoidea, (d) the sea lilies belonging to class Crinoidea, (d) sea anemone
78. The chordates are named for the _____, which is a flexible, rod-shaped structure that is found in the embryonic stage of all chordates and in the adult stage of some chordate species (notochord)

79. In organisms that live in aquatic environments, _____ allow for the exit of water that enters the mouth during feeding (pharyngeal slits)

80. Tetrapod means “four-footed,” and this group includes (amphibians, reptiles, birds, and mammals).

81. In addition to the vertebrates, the phylum Chordata contains two clades of invertebrates which are (Urochordata(tunicates) and Cephalochordata (lancelets).)

82. The best-known extinct vertebrates are the (dinosaurs)

83. Examples of critically endangered vertebrate species include the following, except (a) the Siberian tiger (*Panthera tigris altaica*) (b) the golden frog (*Atelopus zeteki*) (c) the eagle (d) the Dog

84. Jawless fishes are _____ (which includes all the chordate groups except the tunicates and lancelets) that represent an ancient vertebrate lineage that arose over one half-billion years (craniates)

85. Sharks, together with most fishes, also have a sense organ called the _____, which is used to detect movement and vibration in the surrounding water, and a sense that is often considered homologous to “hearing” in terrestrial vertebrates. (lateral line)

86. Nearly all bony fishes have an ossified skeleton with specialized bone cells called _____ that produce and maintain a calcium phosphate matrix. (osteocytes)

87. Bony fishes are further divided into two clades with living members called _____ and _____ (Actinopterygii (ray-finnedfishes) and Sarcopterygii (lobe-finned fishes).)

88. Class Reptilia includes diverse species classified into four living clades. These are the (Crocodilia, Sphenodontia, Squamata, and Testudines).

89. _____ are vertebrates that have hair and mammary glands used to provide nutrition for their young. (Mammals)

90. Mammary hair also serves as a sensory mechanism through specialized hairs called _____, better known as whiskers. (vibrissae)

91. _____ produce milk that is used to feed newborns. (Mammary glands)

92. .Most mammals have _____, meaning they have different types and shapes of teeth that allow them to feed on different kinds of foods. (heterodont teeth)

93. Most mammals are also _____, meaning they have two sets of teeth in their lifetime deciduous or “baby” teeth, and permanent teeth. (diphyodonts)
94. Modern mammals are divided into three broad groups which are _____, _____ and _____: {(monotremes, marsupials, and eutherians (or placental mammals))}
95. Flowerless and seedless plants are also known as _____ [cryptogams]
96. Lower and more primitive plants includes _____, _____ and _____ [thallophytes, bryophytes and pteridophytes]
97. Adaptation deals with the _____, _____ or _____ characteristics that enables a living organism to survive and reproduce in a given environment [physiological, structural, behavioral]
98. Hydrophytes are also known as _____ [water loving plants]
99. The aerenchyma in hydrophytes is important for _____ and _____ [gaseous exchange and buoyancy]
100. Plants adapted to regions with scant or no water and hot conditions are _____ [xerophytes]
101. Xerophytic plants possess _____ to prevent water loss [thick cuticle]
102. _____ possess special ecological adaptation to reduced light intensity [sciophytes]
103. Shade-loving plants possess _____ that reflects outgoing light back into the mesophyll. [red abaxial layer]
104. Sun-loving plants are also known as _____ [Heliophytes]
105. _____ are plants that can tolerate growing under saline conditions [Halophytes]
106. Epiphytes are also known as _____ [air plants/tree dwellers]
107. The mesophyll in epiphytic plants is commonly differentiated into _____ and _____ [hypodermis and chlorenchyma]
108. Lithophytes are _____ [plants with special adaptation to growing on rocks or in rocky terrains having scanty humus]
109. Metalophytes are _____ [plants with special adaptation to natural habitats with toxic level of metals]

110. The term higher plant is used for the group of plants that belongs to the division [spermatophyte]
111. Higher plants are characterized by the production of _____ [seeds]
112. _____ is a special structure employed for the reproduction of higher plants [seeds]
113. Higher plants are characterized by the alternation of _____ [generation]
114. Spermatophyta is subdivided into _____ and _____ [gymnosperm, angiosperm]
115. _____ are less highly developed of the two main sub-divisions of spermatophytes [gymnosperm]
116. There are _____ main classes of gymnosperms [three]
117. The female gametophytes of gymnosperms bear eggs in _____ [archegonia]
118. Gymnosperms have a _____ dominant life cycle [sporophyte]
119. In gymnosperms microspores and megaspores are produced in _____ and _____ respectively [pollen cones, ovulate cones]
120. Angiosperms consists of at least _____ living species classified in _____ families. [260,000 and 453]
121. In angiosperms the ovules are enclosed in a _____ [carpel]
122. In angiosperms, double fertilization leads to the formation of _____ [endosperm]
123. Phloem tissue in angiosperms is composed of _____ and _____ [sieve tube, companion cell]
124. The largest group of angiosperms are [dicots]
125. In angiosperms, double fertilization results in _____ and _____ [zygote, endosperm]
126. The division of an organism into identical left and right halves in only one plane is termed [bilateral symmetry]

127. The division of an organism into two identical halves from any plane passing through the central axis of its body is termed [radial symmetry]

128. In an earthworm, the skin or body wall around the 14th to the 16th segment is called [clitellum]

129. The first metamere of an earthworm where the mouth is situated ventrally is called _____. [peristomium]

130. In arthropods, the body is segmented into _____ and _____ [cephalothorax, posterior abdomen]

131. The body of a snail consists of _____, _____, _____ and _____ [head, mantle, foot, visceral mass]

132. In a starfish, the upper surface where the anus is present is called _____ [aboral surface]

133. The body of a fish is divided into _____, _____ and _____ [head, trunk, tail]

134. In a true fish, unpaired fins are _____, _____ and _____ while paired fins are _____ and _____ [(dorsal, ventral, caudal), (pectoral, pelvic)]

135. In a fish, _____ pairs of gill slits are present laterally between _____ and _____ [five, mouth, pectoral fin]

136. In a frog, several rows of spiny teeth are present in the _____ and absent in the _____ [upper jaw, lower jaw]

137. In a frog, the forelimbs have _____ clawless digits and the hind limbs have _____ clawless digits [four, five]

138. For lizards, the body is divided into _____, _____, _____ and _____ [head, neck, trunk, elongated tail]

139. Lizards possess two pairs of limbs that have _____ clawed digits [five(5)]

140. Perennia plants may be woody or herbaceous [true/false]

141. In Aves, their streamlined body is divided into _____, _____ and _____, [head, neck, trunk]

142. In Aves, the forelimbs are modified into _____ and hind limbs have _____ clawed digits [wings, 4]

143. plants are _____ organisms adapted for life on land [multicellular photosynthetic]

144. In order for plants to be adapted to life on land, they are differentiated into _____ and _____ [hold fast and upper photosynthetic area]

145. Divisions in plant kingdom is _____ in number, it includes _____ bryophytes, _____ seedless plants, _____ gymnosperm and _____ angiosperm [12, 3, 4, 4, 1]

146. The three main groups of flowerless and seedless plants are _____, _____ and _____ [Thallophyta, Bryophyta, Pteridophyta]

147. _____ are the amphibians and pioneers of the plant kingdom [bryophytes]

148. In pteridophytes, _____ contains the reproductive spores [sporangia]

149. Organisms that live rooted to one spot are _____ while organisms that moves around are _____ [sessile, motile]

150. In a poriferan, the mouth is called [ostia]

151. In sponges, the anus is called _____ [osculum]

152. _____ have only one opening functioning as both mouth and anus [coelenterates or cnidarians]

153. Fish, snake and crocodile can be grouped together for possessing [scales]

154. In body symmetry, segmentation is also called _____ [metamerism]

155. _____ is used for defense and food capture in coelenterates [Cnidoblast]

156. The anterior sucker in a platyhelminth acts as an auctorial organ for _____ and _____ [adhesion, ingestion]

157. Anus is present in a tape worm (true/false) [false]

158. In annelids, _____ is a chitinous structure about the middle of each segment which helps in locomotion and food collection [setae or chaetae]

159. In annelids, the genital or copulative papillae are in the _____ and _____ segments [17th, 18th]

160. in echinoderms, _____ is a sieve-like structure that regulate the amount of water in the water vascular system [madreporite]

161. A marine fish with cartilaginous endoskeleton is called a _____, the exoskeleton has _____ scales [Chondrichthyes, placoid]

162. A marine or fresh water fish with a bony endoskeleton is called an _____, the exoskeleton has _____ or _____ scales [Osteichthyes, cycloid, ctenoid]

163. Amphibians do not possess exoskeleton (true/false) [true]

164. Reptiles are without skin gland (true/false) [true]

165. In Aves, _____ serves as exoskeleton and jaws are modified into _____ [feather, beak]

166. In Aves, the eyes consist of three eyelids, they are _____, _____ and _____ [upper, lower, nictitating membrane]

167. Mammals have _____ as exoskeleton [hair]

168. _____ is where the basic needs of the organism to survive are met [Ecosystem]

169. _____ makes up more than half of all vertebrate species [Pisces]

170. Platyhelminthes are divided into three classes namely _____, _____ and _____ [turbellaria, Trematoda, Cestoda]

171. The body cavity of nematodes is considered as a [Pseudo-coelom]

172. In Annelids, excretion and osmoregulation are achieved by ciliate tubules called [nephridia]

173. _____ are the most successive group of animals [arthropods]

174. The body of an arthropod is covered by hard exoskeleton made of _____ [chitin]

175. The shedding of exoskeleton in arthropods is known as _____ [moulting/ecdysis]

176. In arthropods, the compound eyes are made up of several photoreceptor sub-units called _____ [ommatidia]

177. In Arthropods, the body cavity is filled with a fluid called _____, such body cavity is known as _____ [Haemolymph, Haemocoel]

178. _____ are considered to be the second largest group of animals with regards to specie number [Mollusca]

179. In Molluscs, respiration occurs through [gills/ctenidia]
180. The smallest vertebrate is a fish known as _____ [the Philippine goby]
181. The phylum Chordata is sub-divided into four sub-phyla namely , _____, _____ and _____ [hemichordate, urochordate, cephalochordate, vertebrata]
182. The Pisces belongs to the phylum _____ and sub-phylum _____ [Chordata, vertebrata]
183. Fishes maintain a constant internal body temperature [true/false][false]
184. A true fish possesses _____ and _____ [backbone and fin]
185. Pectoral and pelvic fins are used for _____, _____ and _____ [steering, balance and braking]
186. _____ fins keep a fish from rolling over in water [dorsal fin]
187. A Fastswimming fish have _____ tails that provides the thrust needed to speed through water [narrow forked]
188. Slowswimming fish have a _____ tail that helps them swim around rocks or reefs and catch prey [widesquare-shaped]
189. _____ is a flexible bony plate that protects the sensitive gills [operculum]
190. _____ help fishes to move up and down in water [swim bladders]
191. Amphibians have limbs that evolved from _____ and _____ [pectoral and pelvic fins]
192. _____ are the only vertebrates to undergo complete metamorphosis [amphibians]
193. The skin of a frog contains both _____ and _____ [mucus, poisonous gland]
194. A frog has _____ chambered heart with _____ atria and _____ ventricle [3, 2, 1]
195. "systematics" originates from the Greek word _____ which means _____ [systema, placing together]

196. the systematics means _____ of living things in accordance with their _____
(classification, natural relationship)

198. phylum protozoa consist of four classes namey _____, _____, _____ and _____
[Rhizopoda, Ciliata, Fagelata, Sporozoa]

199. _____ is the most important taxon that represents a natural unit
[specie]

200. Single celled organisms are called _____ while multicellular celled organisms are called _____ [protozoans, metazoans]

201. Multicellular organisms without tissue grade are called _____ while those with tissue grades are called _____ [parazoans, eumetazoans]

202. Eumetazoa is sub-divided into two groups namey _____ and _____
[diploblastic, triploblastic]

203. Diploblastic animals have _____ and _____ as the two layers in the body wall [endoderm, ectoderm]

204. Triploblastic animas have _____, _____ and _____ as three layers in the body wall [ectoderm, mesoderm, endoderm]

205. For protozoans, nutrition is either _____ or _____ [autotrophic, heterotrophic]

206. The body wall of a poriferan (sponges) contains _____ [spicules]

207. In coelenterates (cnidarians), the endoderm is separated from the ectoderm by _____ [non-cellular mesoglea]

208. Coelenterates are divided into three classes namely _____, _____ and _____ [Hydrozoa, Scyphozoa, Anthozoa]

209. In Patyhelminthes, _____ is responsible for excretion and osmoregulation [flame cell]

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